🐳 Docker — Step 4

Multi-Tier Apps: a Next.js Frontend + a Node API + Redis, all wired with Compose

Project: `step4` • An Emoji Reaction Board (Next.js → Express → Redis) • Updated 26 June 2026 •
WORK IN PROGRESS

About this guide. This covers what you learned in Stage 4 *so far* — building a three-tier app and making its containers talk to each other. It is intentionally a living document: there's more research and practice still to come, and the last section maps out that road. Everything here is beginner-first: concepts explained from scratch, with the real errors you hit and why they happened.

UPDATED Now includes **per-project env files** (§7) and **WATCHPACK_POLLING / live reload in Docker** (§8).

📋 **Contents**

1. The Big Picture — what Stage 4 adds

Step 3 had two containers (app + Redis). **Step 4 adds a real frontend as its own service**, turning this into a classic **three-tier app**:

```
Browser ──▶ frontend (Next.js) ──▶ backend (Express) ──▶ redis
on your Mac    port 3000              port 4000          port 6379
               └──────── all three on Docker's private network ────────┘
```

- **frontend** — Next.js: what the user sees and clicks (the UI).
- **backend** — Express: a JSON API. No HTML — just data.
- **redis** — stores the data so it survives restarts.

> 💡 **Why three tiers?** Real apps separate the *presentation* (frontend), the *logic/data access* (API), and the *storage* (database). Each can be built, scaled, and restarted on its own. Step 4 is that separation in miniature.

2. The #1 concept: the browser is NOT on the Docker network

This is the single most important idea in Stage 4. Containers can talk to each other by **service name** (e.g. `backend`, `redis`). But your **browser runs on your Mac** — it is *outside* Docker's private network, so it **cannot** reach `http://backend:4000`.

> 🚨 **The trap:** beginners try to call the API from browser code using the service name and it fails — because only *containers* can resolve service names, not the browser.

How Next.js solves it: the Backend-For-Frontend (BFF) pattern

Next.js has **two halves**: a *server* (runs in the container) and the *browser* code it sends out. We route everything through the server:

```
1. Browser calls /api/reactions  ←  same-origin, no CORS needed
2. Next.js server (in the container, ON the network) receives it
3. The server calls http://backend:4000  ←  service name works here!
4. backend asks redis, answer flows back up
```

> 💡 **In one line:** the browser only ever talks to Next.js; the Next.js *server* is the bridge that reaches the API over the Docker network. That's the BFF pattern, and it's how real Next.js apps work.

3. The services & files

```
step4/
├── docker-compose.yml        # wires the 3 services together
├── backend/                  # the Express JSON API
│   ├── .env                  # backend's OWN config (REDIS_URL)
│   ├── index.js              # GET/POST endpoints → Redis
│   ├── package.json
│   └── Dockerfile            # node:20-alpine
└── frontend/                 # the Next.js app
    ├── .env                  # frontend's OWN config (API_URL)
    ├── src/app/page.js               # the UI (runs in browser)
    ├── src/app/api/reactions/route.js# the SERVER bridge to backend
    ├── package.json
    └── Dockerfile            # node:20-slim (see §6 for why)
```

docker-compose.yml (the heart of it)

```
services:
  frontend:
    build: ./frontend
    ports: ["3000:3000"]            # browser reaches Next.js here
    env_file: ["./frontend/.env"]   # frontend's own config (see §7)
    environment:
      WATCHPACK_POLLING: "true"     # reliable hot reload (see §8)
    depends_on: [backend]
    volumes:
      - ./frontend:/app             # live code
      - /app/node_modules           # protect container's modules
      - /app/.next                  # protect Next's build cache

  backend:
    build: ./backend
    env_file: ["./backend/.env"]    # backend's own config (see §7)
    command: npm run dev            # nodemon live reload
    depends_on: [redis]
    volumes:
      - ./backend:/app
      - /app/node_modules
    # no published port — only the Next.js server calls it, internally

  redis:
    image: redis:7
    volumes: ["redis-data:/data"]   # persistence

volumes:
  redis-data:
```

⚠️ **Service name must match.** The service is called `backend`, so the API address must be `http://backend:4000` — not `http://api:4000`. A mismatch gives `ENOTFOUND backend`. (This was a real bug we fixed.)

4. Service-to-service communication

Inside Docker's private network, every service is reachable by its **name** as if it were a hostname. Docker runs an internal DNS that turns the name into the right address.

```
// frontend/src/app/api/reactions/route.js  — runs on the SERVER
const apiUrl = process.env.API_URL || "http://backend:4000";

export async function GET() {
  const res = await fetch(`${apiUrl}/reactions`, { cache: "no-store" });
  return Response.json(await res.json());
}
```

Who calls whom	Address used	Why it works
browser → Next.js	`/api/reactions`	same origin (same site), no network hop outside
Next.js server → backend	`http://backend:4000`	service name on the Docker network
backend → redis	`redis://redis:6379`	service name on the Docker network

5. `RUN` vs `CMD` (a bug you debugged)

These two Dockerfile keywords run at **completely different times** — confusing them caused a real error for you.

	`RUN`	`CMD`
When	while **building** the image	when a **container starts**
Purpose	prepare the image (e.g. `npm install`)	run your app
Can be overridden?	no (already happened)	yes — via compose `command:`

🚨 **The bug:** writing `RUN ["node","index.js"]` tried to start the server *during the build* — before Redis existed — giving `ENOTFOUND redis` and a long hang. The fix: it must be `CMD`, so

the app starts at container-run time when Redis is up.

💡 **Rule of thumb:** `RUN` = build the image. `CMD` = run the app. Tip: if the build log prints `RUN ["node",...]`, your file still says RUN — a correct `CMD` never appears as a build step.

6. Base images: Alpine vs Slim (the SWC saga)

The word after `node:20-` picks which Linux your image is built on. This caused a real headache with Next.js.

Image	Linux	C library	Size	Best for
`node:20`	Debian (full)	glibc	~1 GB	rarely (too big)
`node:20-slim`	Debian (stripped)	**glibc**	~200 MB	**Next.js**
`node:20-alpine`	Alpine	musl	~130 MB	plain Node (your backend)

What went wrong on Alpine

Next.js compiles your code with **SWC**, a *native binary* (built per-OS). Alpine uses **musl** instead of the standard **glibc**, so `npm install` often fails to install the right SWC binary. Next.js then tries to **download it at runtime** — slow and flaky:

```
Downloading swc package @next/swc-linux-x64-musl... to /root/.cache/next-swc
```

⚠️ **Why it never finished:** that download lands in `/root/.cache` — the container's throwaway layer, not a volume. Every rebuild starts a fresh container with an empty cache, so it re-downloads from zero each time.

💡 **The fix:** use `node:20-slim` for the frontend. With glibc, the SWC binary installs cleanly *during the build* (baked into the image) — no runtime download at all. Keep `node:20-alpine` for the plain-Node backend; it has no such native-binary drama.

7. `.env` — who sees what, & two ways to organize it

The `.env` system works in **two layers**, and this trips people up:

1. **Compose reads a `.env`** to fill `${...}` placeholders in `docker-compose.yml`. At this point nothing is inside any container yet.
2. **A variable reaches a container only if it's given to that service** (via `environment:` or `env_file:`). It is per-service — containers do *not* automatically share variables.

Variable	frontend sees it?	backend sees it?
`API_URL`	✅ (given to frontend)	❌
`REDIS_URL`	❌	✅ (given to backend)

UPDATED Two ways to organize your env files

There are two valid setups. We started with the first, then switched to the second.

Way A — one central `.env` + `${substitution}` (a single file next to `docker-compose.yml`):

```
# .env  (project root)
API_URL=http://backend:4000
REDIS_URL=redis://redis:6379
```

```
# docker-compose.yml
  frontend:
    environment:
      API_URL: ${API_URL}        # value pulled from the central .env
  backend:
    environment:
      REDIS_URL: ${REDIS_URL}
```

Way B — a separate `.env` per project + `env_file:` *(what we use now)*:

```
# frontend/.env
API_URL=http://backend:4000

# backend/.env
REDIS_URL=redis://redis:6379
```

```
# docker-compose.yml
  frontend:
    env_file:
      - ./frontend/.env          # loads ALL of frontend's vars into it
  backend:
    env_file:
      - ./backend/.env           # loads ALL of backend's vars into it
```

	Way A — central .env	Way B — per-project .env
Files	one shared `.env`	one `.env` inside each project
Wiring	`environment: KEY: ${KEY}`	`env_file: - ./proj/.env`
Best when	small project, shared config	projects are independent / may be split out
Ownership	everything in one place	each project owns its config (clearer)

> 💡 **Why we switched to Way B:** each project becomes self-contained — its config travels with it, and a service only gets the env file you point it at. Cleaner for multi-project setups. (You only need ONE method; mixing both just duplicates work.)

The Next.js browser rule (still applies either way)

> 🚨 **Next.js gotcha:** only the **server** can read `process.env.API_URL`. The **browser** can't — unless the name starts with `NEXT_PUBLIC_`. That's a safety feature so secrets don't leak to the browser. And `NEXT_PUBLIC_` vars are **baked in at build time**, so changing one needs a rebuild, not just a restart.

> ⚠️ **Changing any env value needs a recreate.** Env vars are set when a container *starts*, not hot-reloaded. After editing an env file, run `docker compose up -d` (it recreates the affected service). A plain `restart` won't pick up new values.

8. NEW Live reload in Docker: `WATCHPACK_POLLING`

You'll see this line in the frontend service:

```
environment:
  WATCHPACK_POLLING: "true"
```

It's a small setting that quietly makes **hot reload actually work** inside Docker on a Mac. Without it, you'd edit a file, save, and... nothing happens in the browser.

The problem it solves

Your code lives on your **Mac**, but Next.js runs inside a **Linux container**, connected by the bind mount (`./frontend:/app`). The file *contents* update fine across that boundary — but the "a file changed!" **notification** usually does not.

```
You edit page.js on the Mac
    │
    ▼
The file IS updated inside the container ✅
    │
    ▼
BUT the change EVENT gets lost crossing Mac → VM → container ❌
    │
    ▼
Next.js never hears about it → NO hot reload 😞
```

On macOS, Docker runs Linux inside a **virtual machine**, and file-change events frequently don't survive the trip into the container.

How polling fixes it (push vs pull)

	OS notifications (default)	Polling (WATCHPACK_POLLING)
How	**push** — the OS tells the watcher "it changed"	**pull** — the watcher keeps re-checking files itself
Speed	instant	slight delay (one poll interval)
CPU	very low	a bit higher (constant checking)
Works across Docker/VM?	**no** (events get lost)	**yes** (doesn't depend on events)

"Checking" = the watcher repeatedly reads each file's **timestamp and size** and compares to last time. Setting `WATCHPACK_POLLING: "true"` switches Next.js's watcher into this pull mode, so it notices your edits regardless of the lost events.

> 🔔 **Analogy:** OS notifications = a doorbell (you relax until it rings — but only if the wiring works). Polling = walking to the door every few seconds to check (tiring, but works even with a broken doorbell). In Docker-on-Mac the doorbell wiring is broken, so we walk.

Why it lives in `environment:`, not in `.env`: it's a *Docker-runtime/dev* tweak, not *application* config. Your app code never reads it — it only matters because you're running in Docker on a Mac. Keeping infra tweaks separate from app config keeps each file focused.

> ⚠️ **Trade-off:** polling uses slightly more CPU (it never sleeps). That's why it's opt-in, not the default. For a learning project the cost is negligible and worth reliable reload.
>
> **Backend note:** nodemon usually handles this fine; if backend changes ever stop reloading, its equivalent is `nodemon --legacy-watch` (its "use polling" switch).

9. Installing libraries: restart vs rebuild

Does `docker compose restart frontend` install a new library? **No.** `restart` only restarts the process — it never runs `npm install` or rebuilds.

What you do	New library works?
`docker compose restart frontend` (only)	❌ nothing installed
`npm install X` on your **Mac** + restart	❌ hidden by the anonymous volume (see below)
`docker compose exec frontend npm install X`	✅ installed in the container's volume
add to `package.json` → `up --build`	✅ baked into the image (permanent)

🚨 **The volume trap.** Because of `- /app/node_modules`, the container's modules live in an anonymous volume that *hides* your Mac's `node_modules`. So installing on your Mac does nothing for the container — the install must happen **inside the container** (`exec ... npm install`) or via a **rebuild**.

💡 **Two clean workflows:** (A) quick dev add → `docker compose exec frontend npm install X`, then `restart frontend` if Next doesn't pick it up. (B) permanent → ensure it's in `package.json`, then `docker compose up --build`. Do (A) for speed, (B) now and then to keep the image in sync.

10. What we built: the Emoji Reaction Board

An interactive board where clicking an emoji bumps a live counter — demonstrating the full chain with a real **POST** (not just GET).

```
app.use(express.json());          // parse JSON request bodies

// All counts live in ONE Redis HASH: reactions = { fire: 12, love: 30, ... }
app.get("/reactions", async (req, res) => {
  const raw = await redisClient.hGetAll("reactions");
  const counts = {};
  for (const k of ["fire","love","wow","nice"]) counts[k] = Number(raw[k]||0);
  res.json(counts);
});

app.post("/reactions", async (req, res) => {
  const { emoji } = req.body;
  if (!["fire","love","wow","nice"].includes(emoji))  // allow-list = validation
    return res.status(400).json({ error: "unknown reaction" });
  const count = await redisClient.hIncrBy("reactions", emoji, 1); // atomic +1
  res.json({ emoji, count });
});
```

New concept	What it taught
Redis **hash** (`hIncrBy` , `hGetAll`)	one key holding many named counters (vs a single integer)
POST with a JSON body	sending data *up* the chain, not just reading
`express.json()`	middleware to parse the request body
Atomic `hIncrBy`	safe even if many people click at the same instant
Allow-list validation	reject unknown input (`banana` → 400 error)
Optimistic UI update	bump the number instantly, then sync to Redis's real value

11. Watching logs in separate terminals

```
docker compose logs -f frontend      # terminal 1
docker compose logs -f backend       # terminal 2
docker compose logs -f redis         # terminal 3
```

- `-f` = **follow** (stream new lines as they happen). `Ctrl+C` stops watching (not the container).
- `--tail 20` limits backlog; `-t` adds timestamps.
- All at once in one terminal: `docker compose logs -f` (color-coded, prefixed).

12. Common errors you hit & their fixes

Error / symptom	Cause	Fix
`ENOTFOUND redis` during **build**	used `RUN` instead of `CMD` — app started at build time	change last line to `CMD ["node","index.js"]`
`ENOTFOUND backend` at runtime	API address didn't match the service name	use `http://backend:4000`
SWC download hangs forever	Alpine/musl needs runtime SWC download	switch frontend to `node:20-slim`
`ECONNRESET` / `npm network aborted`	internet dropped mid-download	retry `up --build` (resumes from cache); restart Docker Desktop
New library "module not found" after `restart`	`restart` doesn't install anything	`exec ... npm install` or `up --build`
Edits don't hot-reload	file-change events lost across Docker/VM	set `WATCHPACK_POLLING: "true"` (§8)
`requires at least 2 arg(s)`	`exec` needs a service *and* a command	`docker compose exec backend sh`

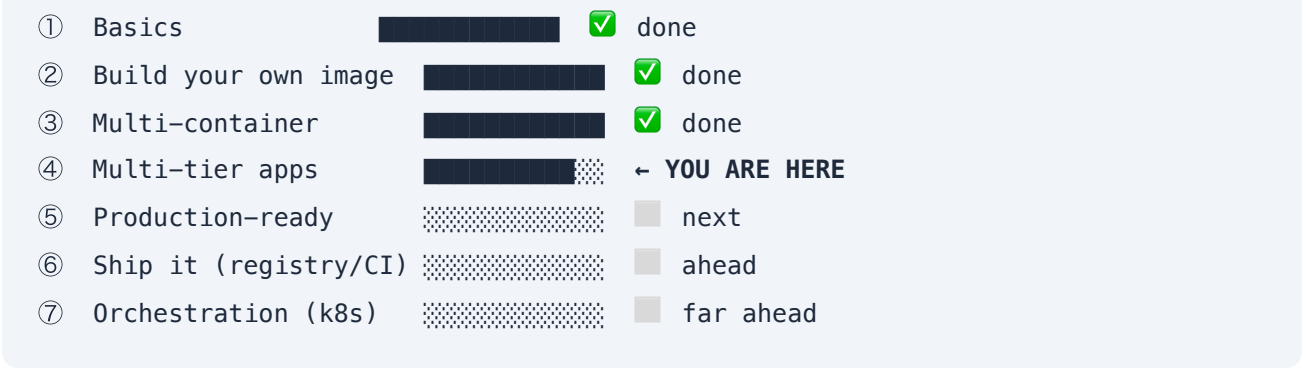
13. Commands cheat-sheet

```
# Lifecycle
docker compose up --build         # rebuild images, then start
docker compose up -d              # start in background (also applies env changes)
docker compose restart frontend   # restart ONE service's process (no install/rebuil
docker compose down               # stop & remove containers + network
docker compose down -v            # ...also remove volumes (wipes Redis data!)

# Inspect
docker compose ps                 # list this project's containers
docker compose logs -f backend    # follow one service's logs

# Inside containers
docker compose exec -it frontend sh             # open a shell
docker compose exec frontend npm install X       # install a lib in the container
docker compose exec redis redis-cli hgetall reactions   # see the Redis hash
```

14. Where you are & the road ahead



You're a solid **intermediate** — you can containerize and run a real multi-service app locally. Roughly **~65–70%** of the Docker a developer needs day-to-day; less of the full production+orchestration world (which most people learn on the job).

Stage 5 — Production-ready (your natural next step)

- **Multi-stage builds** → a tiny production image (Next.js `standalone` output)
- **Healthchecks** + `depends_on: condition: service_healthy` (wait until Redis is truly ready)
- Run as a **non-root user**, restart policies, resource limits
- A **dev/prod compose split** (`docker-compose.override.yml`)

> 💡 **Good news:** Stages 5–6 build on what you already know — they make your current app production-grade, not brand-new concepts. (See also the companion guide on Docker's default network for the networking deep-dive.)

15. Glossary

Term	Meaning
Three-tier app	frontend (UI) + backend (API) + database, as separate services
BFF (Backend-For-Frontend)	the frontend's server makes the API calls on the browser's behalf
Service discovery	reaching a container by its service name as a hostname
Route handler	a Next.js `route.js` that runs on the server (GET/POST)
Client component	React code marked `"use client"` that runs in the browser
`env_file`	loads an entire `.env` file into one service
`NEXT_PUBLIC_`	prefix that exposes a var to the browser (baked in at build)
WATCHPACK_POLLING	makes Next.js watch files by polling — needed in Docker-on-Mac
glibc / musl	the two C libraries; Debian uses glibc, Alpine uses musl
Redis hash	one key holding many named fields/counters
Anonymous volume	unnamed Docker storage used to protect a container-only folder

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